



# HACKTHEBOX



## Tangled Heist

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Prepared By: Nauten, thewildspirit

Challenge Author(s): Nauten

Difficulty: Easy

Classification: Official

## Synopsis

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- Tangled Heist is an easy forensics challenge involving extracting information from a PCAP file and reading the LDAP traffic to answer questions and complete the investigation in order to obtain the flag.

## Description

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- The survivors' group has meticulously planned the mission 'Tangled Heist' for months. In the desolate wasteland, what appears to be an abandoned facility is, in reality, the headquarters of a rebel faction. This faction guards valuable data that could be useful in reaching the vault. Kaila, acting as an undercover agent, successfully infiltrates the facility using a rebel faction member's account and gains access to a critical asset containing invaluable information. This data holds the key to both understanding the rebel faction's organization and advancing the survivors' mission to reach the vault. Can you help her with this task?

## Skills Required

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- Familiarity with network protocol analyzers
- Familiarity with Windows Active Directory structure

## Skills Learned

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- Network analysis
- Analyzing LDAP protocol
- Analyzing KRB5 protocol
- Analyzing Active Directory structure extracting relevant data

## Enumeration

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Players are provided with the following file:

- capture.pcap

The provided file contains network traffic from different protocols, more specifically:

- LDAP traffic of a Windows Active Directory environment (port 389)
- KRB5 (port 88)

The user can parse it with the appropriate tools and extract relevant data needed to reply to questions.

Wireshark has a built-in dissector for LDAP protocol: <https://wiki.wireshark.org/LDAP>

It's really important to understand fully how an Active Directory environment is structured and in which format data is saved in every specific attribute to extract relevant data and reply to questions.

One useful resource is [LDAPWiki](#)

Multiple ways of resolutions are possible, one is to analyze the dump using the Wireshark GUI, alternatively, it's possible to convert the entire dump into a text file using:

```
tshark -r capture.pcap -Y "ldap" -O ldap > ldap_output.txt
```

In this way, the entire dump is readable and can also be parsed.

It's important to notice that multiple streams are available in the pcap file.

## Solution

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### [1/11] Which is the username of the compromised user used to conduct the attack?

We can answer this question by looking at who is authenticated to the LDAP server and sends the rest of the queries.

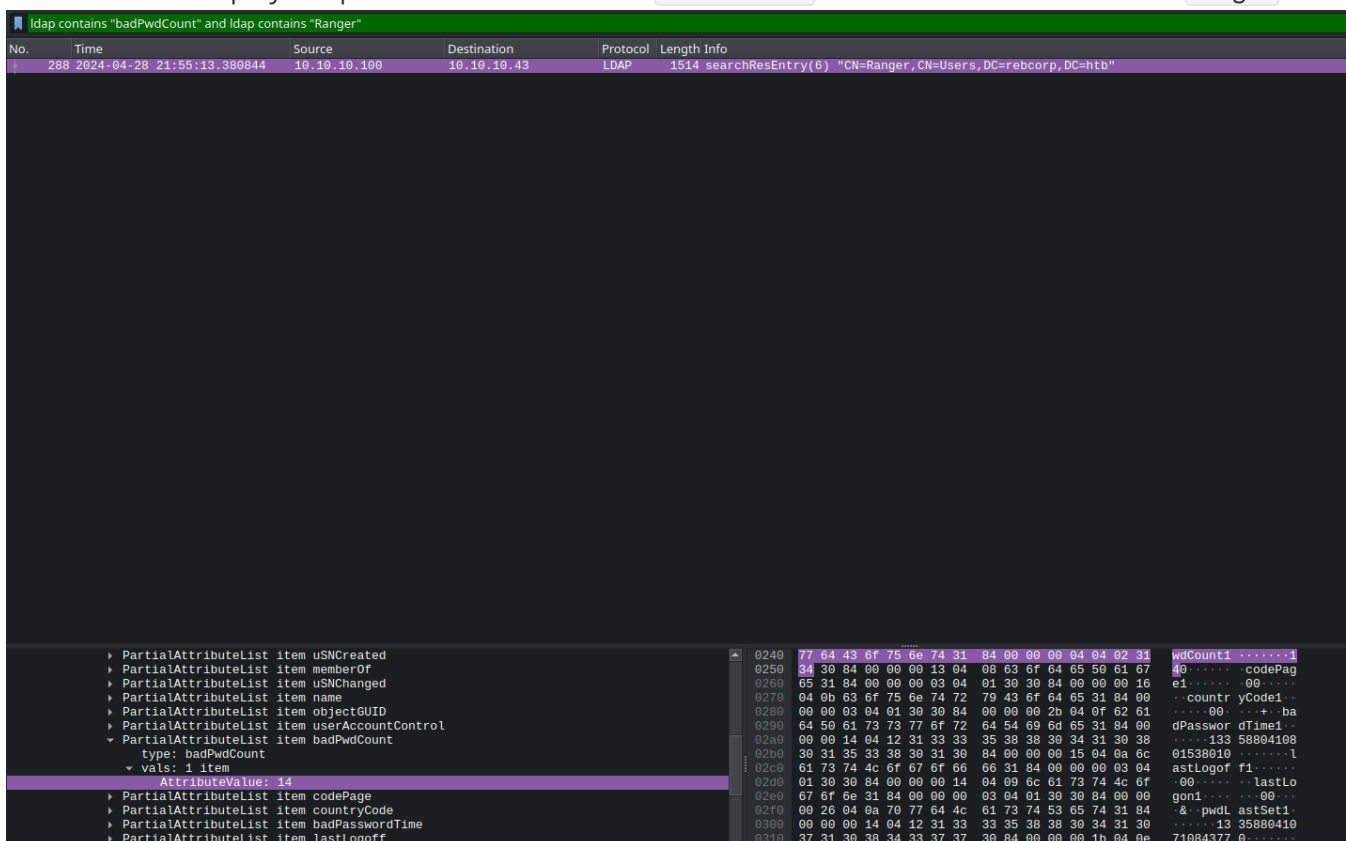


- Answer: `rebcorp.htb`

## [4/11] How many failed login attempts are recorded on the user account named 'Ranger'?

To answer this question we first need to know where such information is stored. After searching on Google, we came across this [attribute](#). As described in Microsoft's official documentation, this attribute indicates the number of times the user tried to log on to the account using an incorrect password. A value of 0 indicates that the value is unknown.

To locate it we will use the following filter: `ldap contains "badPwdCount" and ldap contains "Ranger"`. Wireshark will display the packets that include the `badPwdCount` attribute and the username `Ranger`.



- Answer: `14`

## [5/11] Which LDAP query was executed to find all groups?

Since we know the attacker's IP, we will create a filter that only returns packets sent by the IP `10.10.10.43` and contains the `group` keyword.

| No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Time                       | Source      | Destination  | Protocol | Length | Info                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------|--------------|----------|--------|-----------------------------------------------------|
| 309                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2024-04-28 21:55:13.406066 | 10.10.10.43 | 10.10.10.100 | LDAP     | 160    | searchRequest(8) "DC=reborcorp,DC=htb" wholeSubtree |
| <div> <div> <div>Lightweight Directory Access Protocol</div> <div> <div>LDAPMessage searchRequest(8) "DC=reborcorp,DC=htb" wholeSubtree</div> <div> <div>messageID: 0</div> <div>protocolOp: searchRequest(3)</div> <div>searchRequest <div>baseObject: DC=reborcorp,DC=htb</div> <div>scope: wholeSubtree(2)</div> <div>derefAliases: derefAlways(3)</div> <div>sizeLimit: 0</div> <div>timeLimit: 0</div> <div>typesOnly: False</div> <div>Filter: (objectClass=group)</div> </div> </div> </div> </div> </div> |                            |             |              |          |        |                                                     |

- Answer: (objectClass=group)

## [6/11] How many non-standard groups exist?

We will filter the packets using the "groups" keyword.

| No. | Time                       | Source       | Destination  | Protocol | Length | Info                                                                                                                                                                             |
|-----|----------------------------|--------------|--------------|----------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 276 | 2024-04-28 21:55:13.339618 | 10.10.10.100 | 10.10.10.43  | LDAP     | 947    | searchResEntry(5) "CN=Aggregate,CN=Schema,CN=Configuration,DC=reborcorp,DC=htb"   searchResDone(5) success [80 results]                                                          |
| 309 | 2024-04-28 21:55:13.406066 | 10.10.10.43  | 10.10.10.100 | LDAP     | 160    | searchRequest(8) "DC=reborcorp,DC=htb" wholeSubtree                                                                                                                              |
| 310 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Administrators,CN=Builtin,DC=reborcorp,DC=htb"                                                                                                             |
| 311 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Users,CN=Builtin,DC=reborcorp,DC=htb"                                                                                                                      |
| 312 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Guests,CN=Builtin,DC=reborcorp,DC=htb"                                                                                                                     |
| 313 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Print Operators,CN=Builtin,DC=reborcorp,DC=htb"   searchResEntry(8) "CN=Backup Operators,CN=Builtin,DC=reborcorp,DC=htb"                                   |
| 314 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Replicator,CN=Builtin,DC=reborcorp,DC=htb"                                                                                                                 |
| 315 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Remote Desktop Users,CN=Builtin,DC=reborcorp,DC=htb"   searchResEntry(8) "CN=Network Configuration Operators,CN=Builtin,DC=reborcorp,DC=htb"               |
| 316 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Performance Monitor Users,CN=Builtin,DC=reborcorp,DC=htb"                                                                                                  |
| 317 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Performance Log Users,CN=Builtin,DC=reborcorp,DC=htb"                                                                                                      |
| 318 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Distributed COM Users,CN=Builtin,DC=reborcorp,DC=htb"   searchResEntry(8) "CN=IIS_IUSRS,CN=Builtin,DC=reborcorp,DC=htb"                                    |
| 319 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Cryptographic Operators,CN=Builtin,DC=reborcorp,DC=htb"                                                                                                    |
| 320 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Event Log Readers,CN=Builtin,DC=reborcorp,DC=htb"   searchResEntry(8) "CN=Certificate Service DCOM Access,CN=Builtin,DC=reborcorp,DC=htb"                  |
| 321 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=RDSS Remote Access Servers,CN=Builtin,DC=reborcorp,DC=htb"                                                                                                 |
| 322 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=RDSS Endpoints Servers,CN=Builtin,DC=reborcorp,DC=htb"                                                                                                     |
| 323 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=RDSS Management Servers,CN=Builtin,DC=reborcorp,DC=htb"                                                                                                    |
| 324 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Hyper-V Administrators,CN=Builtin,DC=reborcorp,DC=htb"   searchResEntry(8) "CN=Access Control Assistance Operators,CN=Builtin,DC=reborcorp,DC=htb"         |
| 325 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Remote Management Users,CN=Builtin,DC=reborcorp,DC=htb"                                                                                                    |
| 326 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Storage Replica Administrators,CN=Builtin,DC=reborcorp,DC=htb"                                                                                             |
| 327 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Domain Computers,CN=Users,DC=reborcorp,DC=htb"   searchResEntry(8) "CN=Domain Controllers,CN=Users,DC=reborcorp,DC=htb"                                    |
| 328 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Schema Admins,CN=Users,DC=reborcorp,DC=htb"                                                                                                                |
| 329 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Enterprise Admins,CN=Users,DC=reborcorp,DC=htb"   searchResEntry(8) "CN=Cert Publishers,CN=Users,DC=reborcorp,DC=htb"                                      |
| 330 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Domain Admins,CN=Users,DC=reborcorp,DC=htb"                                                                                                                |
| 331 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Domain Users,CN=Users,DC=reborcorp,DC=htb"                                                                                                                 |
| 332 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Domain Guests,CN=Users,DC=reborcorp,DC=htb"   searchResEntry(8) "CN=Group Policy Creator Owners,CN=Users,DC=reborcorp,DC=htb"                              |
| 333 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=RAS and IAS Servers,CN=Users,DC=reborcorp,DC=htb"                                                                                                          |
| 334 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Server Operators,CN=Builtin,DC=reborcorp,DC=htb"   searchResEntry(8) "CN=Account Operators,CN=Builtin,DC=reborcorp,DC=htb"                                 |
| 335 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Pre-Windows 2000 Compatible Access,CN=Builtin,DC=reborcorp,DC=htb"                                                                                         |
| 336 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Incoming Forest Trust Builders,CN=Builtin,DC=reborcorp,DC=htb"                                                                                             |
| 337 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Windows Authorization Access Group,CN=Builtin,DC=reborcorp,DC=htb"   searchResEntry(8) "CN=Terminal Server License Servers,CN=Builtin,DC=reborcorp,DC=htb" |
| 338 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Allowed RODC Password Replication Group,CN=Users,DC=reborcorp,DC=htb"                                                                                      |
| 339 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Denied RODC Password Replication Group,CN=Users,DC=reborcorp,DC=htb"                                                                                       |
| 340 | 2024-04-28 21:55:13.409635 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Read-only Domain Controllers,CN=Users,DC=reborcorp,DC=htb"                                                                                                 |
| 342 | 2024-04-28 21:55:13.409956 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Enterprise Read-only Domain Controllers,CN=Users,DC=reborcorp,DC=htb"   searchResEntry(8) "CN=Cloneable Domain Controllers,CN=Users,DC=reborcorp,DC=htb"   |
| 343 | 2024-04-28 21:55:13.409956 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Protected Users,CN=Users,DC=reborcorp,DC=htb"                                                                                                              |
| 344 | 2024-04-28 21:55:13.409956 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Key Admins,CN=Users,DC=reborcorp,DC=htb"                                                                                                                   |
| 345 | 2024-04-28 21:55:13.409956 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Enterprise Key Admins,CN=Users,DC=reborcorp,DC=htb"   searchResEntry(8) "CN=Raiders,CN=Users,DC=reborcorp,DC=htb"                                          |
| 346 | 2024-04-28 21:55:13.409956 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Scavengers,CN=Users,DC=reborcorp,DC=htb"                                                                                                                   |
| 347 | 2024-04-28 21:55:13.409956 | 10.10.10.100 | 10.10.10.43  | LDAP     | 1514   | searchResEntry(8) "CN=Agents,CN=Users,DC=reborcorp,DC=htb"   searchResEntry(8) "CN=Watchers,CN=Users,DC=reborcorp,DC=htb"                                                        |
| 349 | 2024-04-28 21:55:13.409956 | 10.10.10.100 | 10.10.10.43  | LDAP     | 903    | searchResEntry(8) "CN=Enclave,CN=Users,DC=reborcorp,DC=htb"   searchResDone(8) success [80 results]                                                                              |
| 667 | 2024-04-28 21:55:16.742845 | 10.10.10.100 | 10.10.10.43  | LDAP     | 947    | searchResEntry(5) "CN=Aggregate,CN=Schema,CN=Configuration,DC=reborcorp,DC=htb"   searchResDone(5) success [3 results]                                                           |

We can see all standard groups, and, at the end of the list, all non-standard groups. Note that on packet 347 groups Agents and watchers are both on the same response.

- Answer: 5

## [7/11] One of the non-standard users is flagged as 'disabled', which is it?

Radiation

To find this information we need to inspect the userAccountControl attribute. The value that is assigned to the attribute tells Windows which options have been enabled.

More details can be found [here](#) and [here](#).

Lightsum field is calculated with the sum of all values. There is also a useful [decoder](#).



In our case, we want: "**NORMAL\_ACCOUNT**" (512) + "**ACCOUNTDISABLE**" (2) = **514**

We can confirm this by using the uac decoder.

## UAC Decoder

Enter the decimal value from the Active Directory "userAccountControl", click "Decode", and see the property flags down below.

Enter the userAccountControl value \*

514

Decode

See the property flags decoded below

|                                                     |                                                    |                                                         |
|-----------------------------------------------------|----------------------------------------------------|---------------------------------------------------------|
| <input type="checkbox"/> SCRIPT                     | <input type="checkbox"/> TEMP_DUPLICATE_ACCOUNT    | <input type="checkbox"/> SMARTCARD_REQUIRED             |
| <input checked="" type="checkbox"/> ACCOUNTDISABLE  | <input checked="" type="checkbox"/> NORMAL_ACCOUNT | <input type="checkbox"/> TRUSTED_FOR_DELEGATION         |
| <input type="checkbox"/> HOMEDIR_REQUIRED           | <input type="checkbox"/> INTERDOMAIN_TRUST_ACCOUNT | <input type="checkbox"/> NOT_DELEGATED                  |
| <input type="checkbox"/> LOCKOUT                    | <input type="checkbox"/> WORKSTATION_TRUST_ACCOUNT | <input type="checkbox"/> USE_DES_KEY_ONLY               |
| <input type="checkbox"/> PASSWD_NOTREQD             | <input type="checkbox"/> SERVER_TRUST_ACCOUNT      | <input type="checkbox"/> DONT_REQ_PREAUTH               |
| <input type="checkbox"/> PASSWD_CANT_CHANGE         | <input type="checkbox"/> DONT_EXPIRE_PASSWORD      | <input type="checkbox"/> PASSWORD_EXPIRED               |
| <input type="checkbox"/> ENCRYPTED_TEXT_PWD_ALLOWED | <input type="checkbox"/> MNS_LOGON_ACCOUNT         | <input type="checkbox"/> TRUSTED_TO_AUTH_FOR_DELEGATION |
|                                                     |                                                    | <input type="checkbox"/> PARTIAL_SECRET_ACCOUNT         |

Searching for the value 514 we find Radiation user as disabled:

```
284 2024-04-28 21:55:13.380534 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(6) "CN=Radiation,CN=Users,DC=rebcorp,DC=htb"
286 2024-04-28 21:55:13.380844 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(6) "CN=Siren,CN=Users,DC=rebcorp,DC=htb"
287 2024-04-28 21:55:13.380844 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(6) "CN=Hawk,CN=Users,DC=rebcorp,DC=htb"
288 2024-04-28 21:55:13.380844 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(6) "CN=Ranger,CN=Users,DC=rebcorp,DC=htb"
289 2024-04-28 21:55:13.380844 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(6) "CN=Gold,CN=Users,DC=rebcorp,DC=htb" | searchResEntry(6)
290 2024-04-28 21:55:13.380844 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(6) "CN=Tiger,CN=Users,DC=rebcorp,DC=htb"
291 2024-04-28 21:55:13.380844 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(6) "CN=Fox,CN=Users,DC=rebcorp,DC=htb"
292 2024-04-28 21:55:13.380844 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(6) "CN=Shadow,CN=Users,DC=rebcorp,DC=htb"
293 2024-04-28 21:55:13.380844 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(6) "CN=Rust,CN=Users,DC=rebcorp,DC=htb"
294 2024-04-28 21:55:13.380844 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(6) "CN=Striker,CN=Users,DC=rebcorp,DC=htb"
295 2024-04-28 21:55:13.380844 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(6) "CN=Scavenger,CN=Users,DC=rebcorp,DC=htb" | searchResEntry(6)
296 2024-04-28 21:55:13.380844 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(6) "CN=Wolf,CN=Users,DC=rebcorp,DC=htb"
297 2024-04-28 21:55:13.380844 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(6) "CN=Steel,CN=Users,DC=rebcorp,DC=htb"
298 2024-04-28 21:55:13.380844 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(6) "CN=Viper,CN=Users,DC=rebcorp,DC=htb"
299 2024-04-28 21:55:13.380844 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(6) "CN=Blade,CN=Users,DC=rebcorp,DC=htb"
300 2024-04-28 21:55:13.380844 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(6) "CN=Wraith,CN=Users,DC=rebcorp,DC=htb" | searchResEntry(6)
301 2024-04-28 21:55:13.380844 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(6) "CN=Hurricane,CN=Users,DC=rebcorp,DC=htb" | searchResEntry(6)
302 2024-04-28 21:55:13.380844 10.10.10.100 10.10.10.43 LDAP 141 searchResRef(6) | searchResDone(6) success [80 results]
304 2024-04-28 21:55:13.402737 10.10.10.100 10.10.10.43 LDAP 187 searchRequest(7) "DC=rebcorp,DC=htb" wholeSubtree
306 2024-04-28 21:55:13.403530 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(7) "CN=SRV195,OU=Domain Controllers,DC=rebcorp,DC=htb" | searchResEntry(7)
307 2024-04-28 21:55:13.403530 10.10.10.100 10.10.10.43 LDAP 142 searchResRef(7) | searchResDone(7) success [80 results]
309 2024-04-28 21:55:13.406066 10.10.10.100 10.10.10.43 LDAP 160 searchRequest(8) "DC=rebcorp,DC=htb" wholeSubtree
310 2024-04-28 21:55:13.409635 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(8) "CN=Administrators,CN=Builtin,DC=rebcorp,DC=htb"
311 2024-04-28 21:55:13.409635 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(8) "CN=Users,CN=Builtin,DC=rebcorp,DC=htb"
312 2024-04-28 21:55:13.409635 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(8) "CN=Guests,CN=Builtin,DC=rebcorp,DC=htb"
313 2024-04-28 21:55:13.409635 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(8) "CN=Print Operators,CN=Builtin,DC=rebcorp,DC=htb" | searchResEntry(8)
314 2024-04-28 21:55:13.409635 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(8) "CN=Replicator,CN=Builtin,DC=rebcorp,DC=htb"
315 2024-04-28 21:55:13.409635 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(8) "CN=Remote Desktop Users,CN=Builtin,DC=rebcorp,DC=htb"
316 2024-04-28 21:55:13.409635 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(8) "CN=Performance Monitor Users,CN=Builtin,DC=rebcorp,DC=htb"
317 2024-04-28 21:55:13.409635 10.10.10.100 10.10.10.43 LDAP 1514 searchResEntry(8) "CN=Performance Log Users,CN=Builtin,DC=rebcorp,DC=htb"

PartialAttributeList item instanceType 0230 74 72 6f 6c 31 84 00 00 00 05 04 03 35 31 34 30 trol1 ..... 5140
PartialAttributeList item whenCreated 0240 84 00 00 00 16 04 00 62 61 64 50 77 64 43 6f 75 ..... b adPwdCou
PartialAttributeList item whenChanged 0250 6e 74 31 84 00 00 00 03 04 01 30 30 84 00 00 00 nt1 ..... 0000
PartialAttributeList item uSNCreated 0260 13 04 08 63 6f 64 65 50 61 67 65 31 84 00 00 00 ..... codeP age1
PartialAttributeList item memberOf 0270 03 04 01 30 30 84 00 00 00 16 04 00 63 6f 75 6e ..... 00 ..... coun
PartialAttributeList item uSNCreated 0280 74 72 70 43 6f 64 65 31 84 00 00 00 03 04 01 30 tryCode1 ..... 0
PartialAttributeList item name 0290 30 84 00 00 00 1a 04 0f 62 61 64 50 61 73 73 77 ..... badPassw
PartialAttributeList item objectGUID 02a0 6f 72 64 54 69 6d 65 31 84 00 00 00 03 04 01 30 ordTime1 ..... 0
PartialAttributeList item userAccountControl 02b0 30 84 00 00 00 15 04 0a 6c 61 73 74 4c 6f 67 6f ..... lastLogo
type: userAccountControl 02c0 66 66 31 84 00 00 00 03 04 01 30 30 84 00 00 00 ffr1 ..... 0000
vals: 1 item 02d0 14 04 09 6c 61 73 74 4c 6f 67 6f 6e 31 84 00 00 ..... lastL ogon1
PartialAttributeList item badPwdCount 02e0 00 03 04 01 30 30 84 00 00 00 26 04 0a 70 77 64 ..... 00 ..... & pwd
type: badPwdCount 02f0 4c 61 73 74 53 65 74 31 84 00 00 00 14 04 12 31 LastSet1 ..... 1
vals: 1 item 0300 33 33 35 38 38 30 34 31 30 36 39 36 37 38 33 38 33588041 06967838
```

- Answer: Radiation

[8/11] The attacker targeted one user writing some data inside a specific field. What is the field name?

We can notice a `modify` request in the LDAP traffic. If we inspect its attributes we can see that the attacker requested to modify the `wwWHomePage`.

|     |            |                 |              |              |      |     |                   |                                                     |
|-----|------------|-----------------|--------------|--------------|------|-----|-------------------|-----------------------------------------------------|
| 669 | 2024-04-28 | 21:55:18.768182 | 10.10.10.43  | 10.10.10.100 | LDAP | 154 | modifyRequest(6)  | "CN=Wraith,CN=Users,DC=rebcorp,DC=htb"              |
| 670 | 2024-04-28 | 21:55:18.769146 | 10.10.10.100 | 10.10.10.43  | LDAP | 76  | modifyResponse(6) | success                                             |
| 671 | 2024-04-28 | 21:55:18.769687 | 10.10.10.43  | 10.10.10.100 | LDAP | 214 | addRequest(7)     | "CN=B4ck,CN=Users,dc=rebcorp,dc=htb"                |
| 672 | 2024-04-28 | 21:55:18.771114 | 10.10.10.100 | 10.10.10.43  | LDAP | 76  | addResponse(7)    | success                                             |
| 673 | 2024-04-28 | 21:55:18.771483 | 10.10.10.43  | 10.10.10.100 | LDAP | 138 | searchRequest(8)  | "cn=Enclave,cn=Users,dc=rebcorp,dc=htb" baseObject  |
| 674 | 2024-04-28 | 21:55:18.771813 | 10.10.10.100 | 10.10.10.43  | LDAP | 192 | searchResEntry(8) | "cn=Enclave,cn=Users,dc=rebcorp,dc=htb" [3 results] |
| 675 | 2024-04-28 | 21:55:18.772165 | 10.10.10.43  | 10.10.10.100 | LDAP | 155 | modifyRequest(9)  | "cn=Enclave,cn=Users,dc=rebcorp,dc=htb"             |
| 676 | 2024-04-28 | 21:55:18.772791 | 10.10.10.100 | 10.10.10.43  | LDAP | 76  | modifyResponse(9) | success                                             |
| 677 | 2024-04-28 | 21:55:18.772939 | 10.10.10.43  | 10.10.10.100 | LDAP | 61  | unbindRequest(10) |                                                     |

|                                                                          |                                                      |                       |
|--------------------------------------------------------------------------|------------------------------------------------------|-----------------------|
| Transmission Control Protocol, Src Port: 49942, Dst Port: 389, Seq: 0000 | 08 00 27 6e f1 3b 08 00 27 45 29 04 08 00 45 00      | ..n;.. 'E)...E..      |
| Lightweight Directory Access Protocol                                    | 0010 00 8c 8a 09 40 00 80 06 00 00 0a 0a 0a 2b 0a 0a | ...@...+...+...       |
| LDAPMessage modifyRequest(6) "CN=Wraith,CN=Users,DC=rebcorp,DC=htb"      | 0020 0a 64 c3 16 01 85 2b 08 1b 97 2f 81 d1 a2 50 18 | ...d...+.../...P...   |
| messageID: 6                                                             | 0030 20 14 29 21 00 00 30 62 02 01 06 66 5d 04 24 43 | ...))!..0b...f]..\$C  |
| protocolOp: modifyRequest (6)                                            | 0040 4e 3d 57 72 61 69 74 68 2c 43 4e 3d 55 73 65 72 | ...N=Wraith ,CN=User  |
| modifyRequest                                                            | 0050 73 2c 44 43 3d 72 65 62 63 6f 72 70 2c 44 43 3d | ...s,DC=reb corp,DC=  |
| object: CN=Wraith,CN=Users,DC=rebcorp,DC=htb                             | 0060 68 74 62 30 35 30 33 0a 01 02 30 2e 04 0b 77 57 | ...htb0503: ...0...ww |
| modification: 1 item                                                     | 0070 57 48 6f 6d 65 50 61 67 65 31 1f 04 1d 68 74 74 | ...WHomePag e1...htt  |
| modification item                                                        | 0080 70 3a 2f 2f 72 65 62 63 6f 72 70 2e 68 74 62 2f | p://rebc orp.htb/     |
| operation: replace (2)                                                   | 0090 71 50 76 41 64 51 2e 70 68 70                   | qPvAdQ.p hp           |
| modification wwWHomePage                                                 |                                                      |                       |
| type: wwWHomePage                                                        |                                                      |                       |
| vals: 1 item                                                             |                                                      |                       |
| AttributeValue: http://rebcorp.htb/qPvAdQ.php                            |                                                      |                       |
| [Response In: 670]                                                       |                                                      |                       |

- Answer: `wwWHomePage`

## [9/11] Which is the new value written in it?

From the previous screenshot, we can also find the new value of the modified attribute.

- Answer: `http://rebcorp.htb/qPvAdQ.php`

## [10/11] The attacker created a new user for persistence. What is the username and the assigned group?

From the traffic, we can notice an `add user` request.

|     |            |                 |              |              |      |     |                   |                                                     |
|-----|------------|-----------------|--------------|--------------|------|-----|-------------------|-----------------------------------------------------|
| 669 | 2024-04-28 | 21:55:18.768182 | 10.10.10.43  | 10.10.10.100 | LDAP | 154 | modifyRequest(6)  | "CN=Wraith,CN=Users,DC=rebcorp,DC=htb"              |
| 670 | 2024-04-28 | 21:55:18.769146 | 10.10.10.100 | 10.10.10.43  | LDAP | 76  | modifyResponse(6) | success                                             |
| 671 | 2024-04-28 | 21:55:18.769687 | 10.10.10.43  | 10.10.10.100 | LDAP | 214 | addRequest(7)     | "CN=B4ck,CN=Users,dc=rebcorp,dc=htb"                |
| 672 | 2024-04-28 | 21:55:18.771114 | 10.10.10.100 | 10.10.10.43  | LDAP | 76  | addResponse(7)    | success                                             |
| 673 | 2024-04-28 | 21:55:18.771483 | 10.10.10.43  | 10.10.10.100 | LDAP | 138 | searchRequest(8)  | "cn=Enclave,cn=Users,dc=rebcorp,dc=htb" baseObject  |
| 674 | 2024-04-28 | 21:55:18.771813 | 10.10.10.100 | 10.10.10.43  | LDAP | 192 | searchResEntry(8) | "cn=Enclave,cn=Users,dc=rebcorp,dc=htb" [3 results] |
| 675 | 2024-04-28 | 21:55:18.772165 | 10.10.10.43  | 10.10.10.100 | LDAP | 155 | modifyRequest(9)  | "cn=Enclave,cn=Users,dc=rebcorp,dc=htb"             |
| 676 | 2024-04-28 | 21:55:18.772791 | 10.10.10.100 | 10.10.10.43  | LDAP | 76  | modifyResponse(9) | success                                             |
| 677 | 2024-04-28 | 21:55:18.772939 | 10.10.10.43  | 10.10.10.100 | LDAP | 61  | unbindRequest(10) |                                                     |

|                                                                          |                                                      |                      |
|--------------------------------------------------------------------------|------------------------------------------------------|----------------------|
| Transmission Control Protocol, Src Port: 49942, Dst Port: 389, Seq: 0000 | 08 00 27 6e f1 3b 08 00 27 45 29 04 08 00 45 00      | ..n;.. 'E)...E..     |
| Lightweight Directory Access Protocol                                    | 0010 00 c8 8a 09 40 00 80 06 00 00 0a 0a 0a 2b 0a 0a | ...@...+...+...      |
| LDAPMessage addRequest(7) "CN=B4ck,CN=Users,dc=rebcorp,dc=htb"           | 0020 0a 64 c3 16 01 85 2b 08 1b fb 2f 81 d1 b8 50 18 | ...d...+.../...P...  |
| messageID: 7                                                             | 0030 20 14 29 5d 00 00 30 81 9d 02 01 07 68 81 97 04 | ...))!..0...h...-    |
| protocolOp: addRequest (8)                                               | 0040 22 43 4e 3d 42 34 63 6b 2c 43 4e 3d 55 73 65 72 | ..."CN=B4ck ,CN=User |
| addRequest                                                               | 0050 73 2c 64 63 3d 72 65 62 63 6f 72 70 2c 64 63 3d | ...s,dc=reb corp,dc= |
| entry: CN=B4ck,CN=Users,dc=rebcorp,dc=htb                                | 0060 68 74 62 30 71 30 18 04 0e 73 41 4d 41 63 63 6f | ...htb0q0...sAMAcco  |
| attributes: 5 items                                                      | 0070 75 6e 74 4e 61 6d 65 31 06 04 04 42 34 63 6b 30 | ...untName1 ...B4ck0 |
| AttributeList item sAMAccountName                                        | 0080 1b 04 11 75 73 65 72 50 72 69 6e 63 69 70 61 6c | ...-userP rincipal   |
| type: sAMAccountName                                                     | 0090 4e 61 6d 65 31 06 04 04 42 34 63 6b 30 13 04 09 | ...Name1... B4ck0... |
| vals: 1 item                                                             | 00a0 67 69 76 65 6e 4e 61 6d 65 31 06 04 04 42 34 63 | ...givenNam e1...B4c |
| AttributeValue: B4ck                                                     | 00b0 6b 30 0c 04 02 73 6e 31 06 04 04 42 34 63 6b 30 | ...k0...sn1 ...B4ck0 |
| AttributeList item userPrincipalName                                     | 00c0 15 04 0b 6f 62 6a 65 63 74 43 6c 61 73 73 31 06 | ...objec tClass1..   |
| type: userPrincipalName                                                  | 00d0 04 04 75 73 65 72                               | ...user              |
| vals: 1 item                                                             |                                                      |                      |

The user is `B4ck`, and then, we notice a `modify` request regarding the group `Enclave`. If we inspect the packet's attributes we can see that the attacker added `B4ck` in the `Enclave` group.

|     |            |                 |              |              |      |     |                   |                                                     |
|-----|------------|-----------------|--------------|--------------|------|-----|-------------------|-----------------------------------------------------|
| 672 | 2024-04-28 | 21:55:18.771483 | 10.10.10.43  | 10.10.10.100 | LDAP | 76  | addResponse(7)    | success                                             |
| 673 | 2024-04-28 | 21:55:18.771813 | 10.10.10.100 | 10.10.10.43  | LDAP | 138 | searchRequest(8)  | "cn=Enclave,cn=Users,dc=rebcorp,dc=htb" baseObject  |
| 674 | 2024-04-28 | 21:55:18.771813 | 10.10.10.100 | 10.10.10.43  | LDAP | 192 | searchResEntry(8) | "cn=Enclave,cn=Users,dc=rebcorp,dc=htb" [3 results] |
| 675 | 2024-04-28 | 21:55:18.772165 | 10.10.10.43  | 10.10.10.100 | LDAP | 155 | modifyRequest(9)  | "cn=Enclave,cn=Users,dc=rebcorp,dc=htb"             |
| 676 | 2024-04-28 | 21:55:18.772791 | 10.10.10.100 | 10.10.10.43  | LDAP | 76  | modifyResponse(9) | success                                             |
| 677 | 2024-04-28 | 21:55:18.772939 | 10.10.10.43  | 10.10.10.100 | LDAP | 61  | unbindRequest(10) |                                                     |

|                                                                                     |                                                      |                       |
|-------------------------------------------------------------------------------------|------------------------------------------------------|-----------------------|
| Transmission Control Protocol, Src Port: 49942, Dst Port: 389, Seq: 1182, Ack: 0000 | 08 00 27 6e f1 3b 08 00 27 45 29 04 08 00 45 00      | ..n;.. 'E)...E..      |
| Lightweight Directory Access Protocol                                               | 0010 00 8d 8a 09 40 00 80 06 00 00 0a 0a 0a 2b 0a 0a | ...@...+...+...       |
| LDAPMessage modifyRequest(9) "cn=Enclave,cn=Users,dc=rebcorp,dc=htb"                | 0020 0a 64 c3 16 01 85 2b 08 1c ef 2f 81 d2 58 50 18 | ...d...+.../...P...   |
| messageID: 9                                                                        | 0030 20 13 29 22 00 00 30 63 02 01 00 66 5e 04 25 63 | ...))!..0c...fa..kc   |
| protocolOp: modifyRequest (6)                                                       | 0040 6e 3d 45 6e 63 6c 61 76 65 2c 63 6e 3d 55 73 65 | ...n=Enclav e,cn=Use  |
| modifyRequest                                                                       | 0050 72 73 2c 64 63 3d 72 65 62 63 6f 72 70 2c 64 63 | ...rs,dc=re bcorp,dc= |
| object: cn=Enclave,cn=Users,dc=rebcorp,dc=htb                                       | 0060 3d 68 74 62 30 35 30 33 0a 01 00 30 2e 04 06 6d | ...=htb0503 ...0...m  |
| modification: 1 item                                                                | 0070 65 6d 62 65 72 31 24 04 22 43 4e 3d 42 34 63 6b | ...ember1S "CN=B4ck   |
| modification item                                                                   | 0080 2c 43 4e 3d 55 73 65 72 73 2c 64 63 3d 72 65 62 | ...CN=User s,dc=reb   |
| operation: add (8)                                                                  | 0090 63 6f 72 70 2c 64 63 3d 68 74 62                | corp,dc= htb          |
| modification member                                                                 |                                                      |                       |
| type: member                                                                        |                                                      |                       |
| vals: 1 item                                                                        |                                                      |                       |
| AttributeValue: CN=B4ck,CN=Users,dc=rebcorp,dc=htb                                  |                                                      |                       |
| [Response In: 676]                                                                  |                                                      |                       |

- Answer: `B4ck,Enclave`

**[11/11] The attacker obtained a hash for the user 'Hurricane' that has the UF\_DONT\_REQUIRE\_PREAUTH flag set. Which is the correspondent plaintext for that hash?**

- Answer: `april18`

For this question, we will extract the hash, convert it in the proper format, and resolve it to plaintext doing the following:

```
$ tshark -r capture.pcap -T pdml > output.pdml
$ krb2john output.pdml > hash
$ john hash --wordlist=/usr/share/wordlists/rockyou.txt
Using default input encoding: UTF-8
Loaded 1 password hash (krb5asrep, Kerberos 5 AS-REP etype 17/18/23 [MD4 HMAC-MD5 RC4 / PBKDF2 HMAC-SHA1 AES 256/256 AVX2 8x])
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
april18 (?)
1g 0:00:00:00 DONE (2024-04-28 16:32) 2.380g/s 14628p/s 14628c/s 14628C/s june10..iheartyou
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
```